

ABBOTSFORD, BC, Canada

WMO: 711080

Lat: 49.027N

Lon: 122.377W

Elev: 59

StdP: 100.62

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.8	-4.4	-17.1	0.8	-4.0	-13.5	1.2	-1.4	12.8	1.7	10.8	3.4	4.2	80	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	30.1	19.6	28.0	18.9	26.1	18.1	20.5	28.5	19.4	26.8	18.4	25.1	3.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	12.4	24.3	16.4	11.7	22.8	15.5	11.1	21.6	59.1	28.8	55.4	26.9	52.2	25.2	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.8	6.5	DB -9.3	33.8	2.5	2.1	-11.1	35.3	-12.6	36.6	-14.0	37.8	-15.9	39.3		
(5)			WB -10.3	22.6	2.5	1.4	-12.1	23.6	-13.6	24.4	-15.0	25.2	-16.9	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	3.8	4.9	7.1	9.9	13.4	15.9	18.6	18.7	15.7	10.7	6.4	3.3		
	DBStd	6.21	3.74	3.11	2.88	2.75	2.94	2.75	2.58	2.36	2.62	2.63	3.46	3.64		
	HDD10.0	815	192	143	97	33	4	0	0	0	0	22	114	208		
	HDD18.3	2863	449	376	349	254	156	84	26	23	86	236	358	466		
	CDD10.0	1084	1	1	7	30	109	177	267	270	171	45	6	1		
	CDD18.3	91	0	0	0	1	2	11	35	35	7	0	0	0		
	CDH23.3	1162	0	0	1	8	54	153	421	400	124	1	0	0		
	CDH26.7	332	0	0	0	1	8	46	134	116	27	0	0	0		
(14) Wind	WSAvg	2.5	2.7	3.0	3.0	2.7	2.4	2.5	2.3	2.1	1.9	2.2	2.7	2.7		
(15) Precipitation	PrecAvg	1168	141	104	102	91	81	63	44	48	72	121	156	145		
	PrecMax	1588	307	272	199	182	177	211	133	169	185	279	290	281		
	PrecMin	712	30	8	24	30	19	16	0	0	5	14	56	19		
	PrecStd	178	62	58	38	32	37	37	31	41	44	60	54	54		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.5	19.1	24.2	27.9	31.0	33.3	32.2	29.5	21.6	15.8	13.6		
		MCWB	10.4	9.3	11.7	14.4	17.3	19.6	21.3	20.3	18.6	14.3	12.9	11.4		
	2%	DB	11.2	12.2	15.7	20.1	24.7	27.7	30.1	29.8	26.7	19.2	13.6	10.5		
		MCWB	8.9	7.7	10.1	12.8	16.0	18.4	20.1	19.7	17.9	13.3	11.5	8.6		
	5%	DB	9.9	10.7	13.4	17.4	22.1	24.5	27.7	27.5	24.2	16.9	12.2	9.0		
		MCWB	7.9	7.4	9.0	11.2	14.9	17.0	19.3	18.9	16.8	12.6	10.2	7.2		
	10%	DB	8.6	9.3	11.7	15.1	19.6	22.0	25.4	25.4	21.8	15.1	11.0	7.7		
		MCWB	6.9	6.7	8.1	10.0	13.9	15.7	18.1	18.1	15.8	12.0	9.2	6.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	10.6	12.8	15.6	18.2	20.7	22.6	21.6	20.0	16.3	13.6	11.4		
		MCDB	12.6	12.3	17.6	22.5	25.8	29.2	31.5	29.8	27.4	19.6	14.9	13.4		
	2%	WB	9.5	9.2	10.9	13.4	16.6	18.7	20.8	20.2	18.6	14.6	12.0	9.0		
		MCDB	10.6	10.8	14.2	18.8	23.2	26.3	28.8	28.2	25.3	17.0	13.3	10.3		
	5%	WB	8.1	8.1	9.7	11.9	15.4	17.2	19.5	19.2	17.4	13.3	10.7	7.6		
		MCDB	9.3	9.7	12.3	16.1	21.1	23.5	26.8	26.5	23.1	15.9	11.8	8.7		
	10%	WB	7.1	7.1	8.7	10.7	14.2	16.0	18.4	18.3	16.3	12.3	9.6	6.4		
		MCDB	8.3	8.9	11.2	14.3	19.0	21.5	24.8	24.8	20.6	14.6	10.7	7.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.7	7.6	8.9	9.6	9.4	11.2	11.5	10.6	7.8	6.0	5.0		
		MCDBR	7.0	9.1	10.9	13.7	14.7	15.0	16.0	16.2	16.0	11.2	7.3	6.2		
	5% WB	MCWBR	5.0	5.7	6.4	7.8	7.9	7.6	7.9	7.9	8.4	6.7	5.1	4.6		
		MCWBR	6.1	7.2	9.4	12.0	13.2	13.6	15.0	14.9	14.3	9.4	6.4	5.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.314	0.309	0.326	0.361	0.388	0.363	0.358	0.361	0.345	0.338	0.319	0.306		
		taud	2.457	2.507	2.477	2.376	2.326	2.411	2.445	2.456	2.496	2.502	2.485	2.444		
	Ebn at Noon	Ebn at Noon	748	852	891	888	873	897	895	877	859	802	735	703		
		Edh at Noon	63	75	90	111	121	112	107	101	88	75	61	56		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.90	1.76	2.67	4.12	5.30	5.71	6.38	5.41	3.78	2.08	1.04	0.71		
	RadStd	RadStd	0.14	0.25	0.34	0.41	0.45	0.64	0.53	0.40	0.33	0.27	0.18	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+20
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page